# SALES TREND AND TIME-BASED PERFORMANCE ANALYSIS

Afficionado Coffee Roasters

A Research Paper on Transaction Volume, Revenue,
Time-of-Day Demand, and Store-Level Performance

**Prepared by**

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Project Report | Exploratory Data Analysis (EDA) | Streamlit Dashboard

Prepared using the available Afficionado Coffee Roasters transaction dataset and project analysis.

## Abstract

This project examines transaction-level sales data from Afficionado Coffee Roasters to understand how sales activity varies by hour of the day and across store locations. The analysis focuses on transaction volume, quantity sold, estimated revenue, time-of-day groups, and location-specific demand patterns. The workflow includes data validation, feature engineering, exploratory data analysis (EDA), grouped aggregations, visual comparisons, and an interactive Streamlit dashboard. Revenue is calculated as transaction quantity multiplied by unit price. Hourly and store-level summaries are used to support operational questions such as when demand is strongest and whether stores share similar demand patterns. A key data limitation was identified: the available dataset contains a year field fixed at 2025 and a transaction-time field, but no actual transaction-date column. Consequently, reliable daily or weekly trends, day-of-week comparisons, and weekday-versus-weekend analysis cannot be established from this version of the data. The report documents this limitation rather than generating artificial dates. The resulting dashboard provides an exploratory view of supported time-of-day and store-level measures and can help inform staffing and operational planning once findings are reviewed alongside business context.



**Keywords:** Retail analytics, coffee sales, exploratory data analysis, transaction volume, revenue, time-of-day analysis, store comparison, Streamlit


## 1. Introduction

Retail businesses need to understand not only what customers purchase, but also when purchases occur and how demand differs between locations. For a coffee business, time-based demand information can support staff allocation, preparation planning, stock availability, and service efficiency. Transaction-level records provide a practical foundation for this analysis because each record contains transaction-related information such as quantity, unit price, product attributes, store location, and transaction time.

This project analyzes the available Afficionado Coffee Roasters dataset to summarize sales activity by hour and time-of-day group and to compare patterns across store locations. The work combines data preparation, calculated measures, visual exploration, and an interactive Streamlit dashboard. The report follows an academic research-paper structure while remaining focused on the analysis actually supported by the available data.

## 2. Problem Definition

The project brief indicates that the business has transaction-level data but needs a consolidated understanding of sales patterns, busy and quiet periods, and location-level differences. Without clear summaries, operational decisions about staffing and store planning may rely on intuition rather than observed transaction activity.

The available dataset, however, does not include a true transaction date. This restricts the analysis: hours can be analyzed because transaction_time is available, but calendar-based trends and weekday/weekend comparisons require a date field. This constraint is treated as a data limitation throughout the report.

## 3. Research Objectives

The primary objectives are to:

- Examine transaction volume, quantity sold, and calculated revenue across available hours.
- Compare demand across defined time-of-day groups.
- Compare transaction activity and revenue across store locations.
- Identify the busiest hour for the selected measure using grouped data.
- Present supported findings through an interactive Streamlit dashboard.

The broader project brief also asks for daily/weekly sales trends and day-of-week analysis. These objectives cannot be completed reliably with the current dataset because the actual transaction date is absent. They should be revisited if a dated version of the source data becomes available.

# 4. Dataset Description

The dataset used for the analysis contains 149,116 rows and 11 original columns. The recorded fields include transaction_id, year, transaction_time, transaction_qty, unit_price, store_id, store_location, product_id, product_category, product_type, and product_detail. The year field contains 2025 for all records. The transaction_time field provides time-of-day information but does not preserve an actual calendar date in the available version.

During preparation, transaction_time was parsed as a time value. As a result, its date component defaults to a placeholder date (1900-01-01). That placeholder is not a real transaction date and must not be used to infer the day of the week or construct calendar-based trends.

# 5. Methodology

## 5.1 Data Ingestion and Validation

The transaction dataset was loaded into a pandas DataFrame in Jupyter Notebook. The data preparation workflow included inspecting column names and data types, checking for missing values and duplicate records, reviewing transaction identifiers, and checking whether quantity and unit-price values were valid for revenue calculations. Any data-quality decisions should be retained in the notebook so that the analysis can be reproduced.

## 5.2 Feature Engineering

A revenue field was calculated as transaction quantity multiplied by unit price: Revenue = transaction_qty × unit_price. The hour was extracted from transaction_time to enable hourly aggregation. A time_bucket field was also created to group records into broad periods: Morning (06:00–11:59), Afternoon (12:00–16:59), Evening (17:00–21:59), and Late hours (22:00–05:59). The dataset's observed operating times are concentrated within the available daytime/evening period, so empty or unobserved hour ranges should not be interpreted as evidence of zero demand outside business operations.

## 5.3 Exploratory Data Analysis

The EDA aggregates records by hour and by time bucket. For each group, transaction count, total quantity, and total revenue are calculated. Store-level analysis groups records by store location and hour to compare hourly transaction activity. A heatmap provides a compact comparison of transaction counts by store and hour, while bar charts summarize location-level activity. These visualizations are descriptive: they reveal patterns in the recorded data but do not by themselves establish the causes of those patterns.

## 5.4 Interactive Dashboard

A Streamlit dashboard was created to present key summary metrics, hourly charts, time-of-day comparisons, a store-by-hour heatmap, and store-level comparisons. Filters allow users to select a store and an hour range, while a metric selector switches the displayed measure between revenue and quantity. The dashboard is intended to make exploratory results easier to inspect and discuss.

# 6. Results and Analysis

## 6.1 Hourly Demand

Hourly transaction analysis is supported by the data because transaction_time is available. The analysis groups transactions by hour and calculates transaction count, quantity sold, and revenue. The busiest hour should be reported from the row with the highest transaction count; the highest-revenue hour should be reported separately because transaction volume and revenue are different measures. The dashboard includes an hourly chart to help users compare these measures across observed hours.

The exact peak-hour values and revenue totals are intentionally not stated in this report because the numerical aggregation output was not included in the material available while preparing this document. They should be copied from the final notebook output or dashboard before formal submission.

## 6.2 Time-of-Day Groups

Grouping transactions into morning, afternoon, evening, and late-hour categories gives a simpler operational view than inspecting every hour separately. The grouped summaries compare transaction counts, quantities, and revenue. These summaries can help management identify which periods warrant closer review for staffing and preparation needs. The period with the most transactions should not automatically be assumed to be the period with the highest revenue; both measures should be checked.

## 6.3 Store-Level Comparison

Store-level aggregation compares total transactions, quantity sold, and revenue across locations. The store-by-hour heatmap supports comparison of the timing of demand, including whether stores appear to share similar peaks or show different hourly patterns. Differences in total volume should be interpreted carefully because stores may differ in operating hours, customer traffic, product mix, and other factors not evaluated in this analysis.

## 6.4 Daily and Weekly Trends

Daily revenue trends, weekly aggregation, average revenue by weekday, and weekday-versus-weekend comparisons require a genuine transaction-date field. The available dataset does not contain this field. Because the date attached during time parsing is only a software placeholder, it is not valid evidence of the actual day on which a transaction occurred. These analyses are therefore reported as unavailable rather than estimated or fabricated.

# 7. Discussion and Business Implications

The analysis and dashboard provide a reusable framework for reviewing demand by hour, broad time-of-day group, and store location. If the hourly results show clear concentrations of transactions, managers can consider aligning staffing and preparation activities with those periods. If stores have different hourly patterns, schedules may need to be adapted by location instead of applying one standard schedule to every store. Such operational decisions should be made after reviewing the exact charts and confirming that observed patterns are consistent across a suitable period.

Revenue and transaction count should be interpreted together. A period with many transactions may have a lower average transaction value, while a period with fewer transactions may generate more revenue if orders are larger or product prices differ. Store comparisons should likewise distinguish overall volume from revenue and quantity sold.

## 8. Recommendations

1. Use the hourly transaction chart to review the periods with the greatest recorded demand and consider whether staffing levels match that demand.
2. Compare transaction count, quantity, and revenue together before changing staffing or inventory practices.
3. Review the store-by-hour heatmap to determine whether locations need different staffing schedules.
4. Validate operational recommendations with store managers and, where possible, compare the patterns over several dated periods.
5. Obtain a source dataset that includes the actual transaction date so daily, weekly, weekday, and weekend patterns can be analyzed correctly.
6. Keep data preparation and calculations documented in the Jupyter Notebook, and use the Streamlit dashboard as the interactive presentation layer.
7. Before submission, add the exact peak transaction hour, peak revenue hour, and relevant numerical totals from the final notebook output. Do not report values that have not been verified.

## 9. Limitations

The most important limitation is the absence of an actual transaction-date column. This prevents reliable analysis of daily and weekly trends and day-of-week patterns. The fixed year value of 2025 does not supply the missing month and day. The placeholder date created while parsing transaction_time cannot be treated as real calendar information. In addition, the present report does not include exact peak-hour totals because the final numeric aggregation outputs were not available during report preparation. The findings are descriptive and do not prove that time or location alone causes differences in sales.

## 10. Conclusion

This project establishes an exploratory approach for understanding when transactions occur at Afficionado Coffee Roasters and how recorded activity differs across store locations. Through data preparation, revenue calculation, hourly and time-bucket aggregation, store-level comparison, and a Streamlit dashboard, the work provides a practical basis for reviewing demand and considering operational planning. The current dataset supports time-of-day and location analysis, but it does not support reliable calendar-based trends because the actual transaction date is missing. Obtaining a dated dataset and inserting verified numerical results would strengthen the report and allow the full set of project objectives to be addressed.

## 11. Executive Summary for Stakeholders

This project organizes coffee-shop transaction data into an interactive view of transaction volume, quantity sold, estimated revenue, time-of-day groups, and store-level demand. The dashboard can help stakeholders explore when activity is concentrated and whether stores show different hourly patterns. Potential operational uses include reviewing staffing coverage, preparation schedules, and location-specific planning. However, the supplied dataset lacks actual transaction dates, so day-of-week and daily/weekly trend claims cannot currently be supported. Stakeholders should treat the dashboard as a descriptive decision-support tool, validate any staffing changes against operational experience, and request a dated source file to complete the calendar-based analysis.

## 12. References and Project Materials

• Unified Mentor project brief: “Sales Trend and Time-Based Performance Analysis for Afficionado Coffee Roasters” (project documentation supplied by the project provider).
• Afficionado Coffee Roasters transaction dataset, as used in the project notebook.
• Streamlit documentation: https://docs.streamlit.io/
• pandas documentation: https://pandas.pydata.org/docs/
• Matplotlib documentation: https://matplotlib.org/stable/
• Seaborn documentation: https://seaborn.pydata.org/

## Appendix A. Final Checks Before Submission

Before submitting this report, update it with verified outputs from the notebook: (a) busiest hour by transaction count, (b) highest-revenue hour, (c) transaction count/revenue/quantity by time bucket, and (d) store-level comparison values. Insert selected screenshots of the hourly chart, time-of-day chart, and store-by-hour heatmap if your submission requirements permit. Ensure the report and dashboard use the same cleaned dataset and calculations.